Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-8

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following does not characterize big data analytics?

    a. Volume
    b. Velocity
    c. Vector
    d. Variability

Answer: (c)

Justification: Please refer to Slide 7 of Lecture 1 of Week 8

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2. Which of the following algorithms is useful when it is feasible to acquire training data for the different states (or classes) that need to be modeled?

    a. Unsupervised learning algorithm
    b. Supervised learning algorithm
    c. Reinforcement learning algorithm
    d. None of the above

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 1 of Week 8

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3. “Which of the following is false about the deep learning models?

    a. Learns features automatically without manually specifying them
    b. Does not use neural networks
    c. Raw data from sensors can be directly fed into the model
    d. Generally require a large amount of computation power

Answer: (b)

Justification: Please refer to Slide 9 of Lecture 1 of Week 8

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4. Which of the following software may be used for batch-oriented analytics?

a. Apache Hadoop and Apache Spark
b. Eclipse
c. Sublime
d. None of the above

Answer: (a)

Justification: Please refer to Slide 13 of Lecture 1 of Week 8

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5. Which of the following is an example of a classification algorithm?

a. Density-Based Spatial Clustering of Applications with Noise
b. Support Vector Machine
c. K-Means
d. Bayes Regression

Answer: (b)

Justification: Please refer to Slide 14 of Lecture 1 of Week 8

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6. In _____, data points may belong to more than one cluster. Fill in the blanks.

a. Hard Clustering
b. Soft Clustering
c. Hard Classification
d. Soft Classification

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 2 of Week 8

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7. Which of the following supervised learning algorithms may be used when the output variable is a real number instead of a categorical value?

a. Regression
b. Classification
c. Both (a) and (b)
d. Neither (a) nor (b)

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 2 of Week 8

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8. Which of the following statements about the reinforcement learning algorithm is false?

    a. There is no external supervisor to guide the agent
    b. Builds a knowledge graph from the constant feedback of the corresponding actions
    c. There is no reward function
    d. There is a mapping between input and output

Answer: (c)

Justification: Please refer to Slide 15 of Lecture 2 of Week 8

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9. “Machine learning is a subset of deep learning that can learn automatically by finding the features of the object on its own.” State true or false.

    a. True
    b. False

Answer: (b)

Justification: Please refer to Slide 3 of Lecture 3 of Week 8

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10. Why is cloud computing necessary for IIoT?

    a. High computational speed for data monitoring and analytics
    b. Scalable and secure services
    c. Effective data acquisition
    d. All of the above

Answer: (d)

Justification: Please refer to Slide 5 of Lecture 4 of Week 8

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11. Which of the following is not true about Platform-as-a-service for IIoT?

    a. Allows industries to self-develop applications
    b. Clients have control over the application and the configuration environment
    c. The service provider manages everything
    d. Software firms like Cumulocity, Bosch IoT, and Carriots offer PaaS for IoT industries

Answer: (c)

Justification: Please refer to Slide 9 of Lecture 4 of Week 8

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12. Who among the following does not provide Industrial Cloud Platform?

    a. C3 IoT
    b. Uptake
    c. Netbeans
    d. GE Predix

Answer: (c)

Justification: Please refer to Slide 16 of Lecture 4 of Week 8

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13. Which of the following is/are the advantage(s) of using a Cloud simulator before real deployment?

    a. Pre-deployment test before real implementation
    b. System testing at no cost
    c. Repeatable evaluation of the system
    d. All of the above

Answer: (d)

Justification: Please refer to Page 246 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy published by Cambridge University Press

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14. Which of the following does not represent the objective of Industry 4.0?

    a. Rigid solutions
    b. Expanded security
    c. Better performance
    d. Higher production

Answer: (a)

Justification: Please refer to Slide 19 of Lecture 5 of Week 8

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15. “K-medoid is a clustering algorithm that is used to maximize the distance between some randomly selected points in a cluster.” State true or false.

    a. True

b. False

Answer: (b)

Justification: Please refer to Page 260 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee published by CRC Press

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